

Counting Technique

Multiplication

$$n_1 \times n_2$$

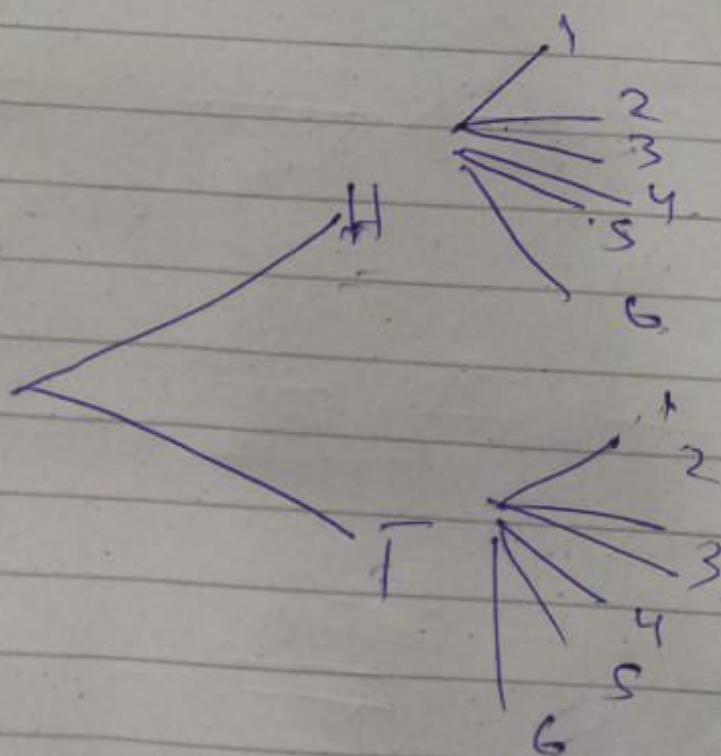
$\Rightarrow n_1 n_2$ ways

Coin and Dice are thrown together

$$2 \times 6 = (12) \text{ ways}$$

outcome $\rightarrow$ sample

$\{ H_1, H_2, H_3, H_4, H_5, H_6$
 $T_1, T_2, T_3, T_4, T_5, T_6 \}$



22-11-20

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1000000

Permutation

arrangement of all rows

abc
abc, acb, bac, baa
cab, cba

$$n! \Rightarrow 3! \Rightarrow 3 \times 2 \times 1 = 6$$

$$0! = 1$$

arrange in circle

$$(n-1)!$$

$$(3-1)! \Rightarrow 2! \quad c \quad b$$

b

$$n \rightarrow S$$

$$6 \rightarrow obj = c$$

5 letter $\rightarrow$ 3 letter

$$R = \frac{n!}{(n-6)!} \Rightarrow \frac{5!}{1}$$

$$SP_3 = 60$$

Combination

Selection

$${}^nC_r = \frac{n!}{(n-r)! r!}$$

5 letters (a b c d e)

3 letters select.

$${}^5C_3 = \frac{5!}{(5-3)! 3!}$$

Using the digit 0, 1, 2, 3, 4, 5, 6 how many 4 digit even numbers can be made, no digit being repeated

0, 2, 4, 6 = five

Unit place 0

$$\frac{4}{1} \times \frac{5}{1} \times \frac{6}{1} \times \frac{1}{1} = 120$$

Unit place 2

$$\frac{5}{1} \times \frac{5}{1} \times \frac{4}{1} \times \frac{1}{1} = 100$$

Unit place 4

$$\frac{5}{1} \times \frac{5}{1} \times \frac{4}{1} \times \frac{1}{1} = 100$$

Unit place 6

$$\frac{5}{1} \times \frac{5}{1} \times \frac{4}{1} \times \frac{1}{1} = 100$$

$$100 + 100 + 100 + 20 = 420$$

1) The no. of permutation of n object arranged in circle is $(n-1)!$.

2) The no. of distinct permutation of n thing of which n_1 are one kind, n_2 of second kind
----- k th kind is.

$$\frac{n!}{n_1! n_2! \dots n_k!}$$

3) In a college football training session the ~~defect~~ ^{defensive} coordinator need to have 10 players standing in a row. Among these 10 players there are 1 Freshman, 2 Sophomores, 4 Juniors and 3 Seniors. How many different way can they be arrange in rows if only their class level be distinguished.

$$\frac{10!}{1! 2! 4! 3!} = 12,600.$$

Counting Rules

2.1 multiplication Rules

operation in n_1 ways and second
operation in n_2 ways.
Together $n_1 n_2$ ways.

) How many sample space when a
pair of dice is thrown once.

) How many even four digit
numbers can be formed if
digit 1, 3, 5, 6 and 9
if each digit can be used
only once.

$$\frac{1}{1} \times \frac{1}{1} \times \frac{1}{1} \times \frac{1}{1} \times \frac{1}{1} \times \frac{1}{1} \times \frac{1}{1} \times \frac{1}{1} \times \frac{1}{1} \times \frac{1}{1}$$

$$1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1$$

$$1 \times 5 \times 4 \times 3 \Rightarrow 60$$

$$2 \times 4 \times 4 \times 3 = 96$$

$$156$$

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--- k th kind is.

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3) In a college football training session the ~~defect~~ defensive coordinator need to have 10 players standing in a row. Among these 10 players there are 1 Freshman, 2 Sophomores, 4 Juniors and 3 Seniors. How many different way can they be arrange in rows if only their class level be distinguished.

$$\frac{10!}{1! 2! 4! 3!} = 12,600$$

Using the digit 0, 1, 2, 3, 4, 5, 6 how many 4 digit even numbers can be made, no digit being repeated

Unit place 0 $\frac{4}{1} \times \frac{5}{1} \times \frac{6}{1} \times \frac{10}{1} \Rightarrow 120$

Unit place 2 $\frac{5}{1} \times \frac{5}{1} \times \frac{4}{1} \times \frac{2}{1} = 100$

$\downarrow$
 $\frac{5}{2,0}$

Unit place 4 $\frac{5}{1} \times \frac{5}{1} \times \frac{4}{1} \times \frac{4}{1} = 100$

$\downarrow$
 $\frac{5}{2,0}$

Unit place 6 $\frac{5}{1} \times \frac{5}{1} \times \frac{4}{1} \times \frac{6}{1} = 100$

$\downarrow$
 $\frac{5}{2,0}$

$100 + 100 + 100 + 20 = 420$

permutation

is arrangement of all object
of a set of object

consider the three letters
(a, b, and c) $\Rightarrow 3 \times 2 \times 1 = 6$
abc, acb, bac, bca, cab, cba

$$n! = n(n-1) \dots 1$$

In general n distinct object
Taken r at a time

$${}_n P_r = \frac{n!}{(n-r)!}$$

In one year, three award
(research, teaching and service)
will be given to a class
of 25 graduate student
in a statistics department
If each student can
receive at most one
award, how many possible
selection are there?

Since award are distinguishable
It is a permutation problem

$${}_{25} P_3 = \frac{25!}{22!} = 13800$$

permutation

is arrangement of all object
of a set of object

consider the three letters
(a, b, c) $\Rightarrow 3 \times 2 \times 1 = 6$
abc, acb, bac, bca, cab, cba

$$n! = n(n-1) \dots 1$$

In general n distinct object
Taken t at a time

$${}_n P_t = \frac{n!}{(n-t)!}$$

In one year, three award
(research, teaching and service)
will be given to a class
of 25 graduate student
in a statistics department
If each student can
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Since award are distinguishable
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$${}_n P_3 = \frac{25!}{22!} = 13800$$

only the 6 stars level be distinguished.

How many different letter arrangement can be made from the letters in the word statistics?

$$\frac{10!}{3!3!2!1!1!1!}$$

$$= 50,400$$

1) The no. of permutation of n object arranged in circle is $(n-1)!$

2) The no. of distinct permutation of n thing of which n_1 are one kind, n_2 of second kind
--- k th kind is.

$$\frac{n!}{n_1! n_2! \dots n_k!}$$

3) In a college football training session the defect 'defensive' combination need to have 10 players standing in a row. Among these 10 players there are 1 Freshman, 2 Sophomores, 4 Juniors and 3 Seniors. How many different way can they be arrange in rows if only their class level be distinguished.

$$\frac{10!}{1! 2! 4! 3!} = 12,600$$

) permutation

$$\frac{n!}{n_1! n_2! n_3!}$$

Statistics

$$\frac{10!}{3! 3! 2!} \Rightarrow \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1 \cdot 3 \cdot 2 \cdot 1 \cdot 2 \cdot 1}$$

aksan

$$\frac{8!}{2!} \Rightarrow \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 1}$$

)

$$\frac{10!}{2! 4! 3!}$$

1605000

Permutation

argument of all 60005

abc

abc, acb, bac, bca

cab, cba

$$n! \Rightarrow 3! \Rightarrow 3 \times 2 \times 1 = \boxed{6}$$

argument in circle

$$(n-1)!$$

0, 1, 2, 3, 4, 5, 6

4 -> Digit even number
can not be 5 no digit
being repeated

$$\frac{6}{1} \times \frac{5}{1} \times \frac{4}{1} \times \frac{1}{0} = 120$$

$$\frac{5}{1} \times \frac{5}{1} \times \frac{4}{2} \times \frac{1}{2} = 100$$

$$\frac{5}{1} \times \frac{5}{1} \times \frac{4}{4} \times \frac{1}{1} = 100$$

$$\frac{5}{1} \times \frac{5}{1} \times \frac{4}{6} \times \frac{1}{6} = 100$$

420

0
2
4
6

~~18~~

1 Pran

0! = 1

>

only these
distinguished.

$$\frac{10!}{1! 2! 4! 3!} = 12,600$$

) In how many can 7 graduate student be assigned to 1 triple and 2 double hotel rooms during a conference?

$$\frac{7!}{3! 2! 2!} = 210$$

No. of ways of selecting r -object from n without regard to order is called combinations

$$\binom{n}{r} = \frac{n!}{(n-r)! r!}$$

A young boy asks his mother to get 5 Game-Boy™ cartridges from his collection of 10 arcade and 5 sport game. How many ways are there that his mother can get 3 arcade and 2 sport game?

$$\binom{10}{3} = 120, \binom{5}{2} = 10$$

using multi rule

$$120 \times 10 = 1200 \text{ way}$$

Counting Rules

Multiplication Rules

operation in n_1 ways and second
operation in n_2 ways.
Together $n_1 n_2$ ways.

How many sample space when a
pair of dice is thrown once.

How many even four digit
numbers can be formed if
digit 1, 3, 5, 6 and 9
if each digit can be used
only once.

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$$

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{16}$$

$$1 \times 5 \times 4 \times 3 = 60$$

$$2 \times 4 \times 4 \times 3 = 96$$

$$156$$